Jeans-Independent Holdback Bypass for AMR Refinement in Cosmological Simulations

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Abstract

Cosmological AMR simulations employ a *holdback* mechanism that progressively releases refinement levels as the universe expands, preventing premature over-refinement at high redshift. However, this mechanism blocks *all* refinement criteria simultaneously, including the Jeans length criterion, which is a numerical stability requirement rather than merely a resolution preference. When a holdback level is finally released, a large number of cells suddenly satisfy the refinement condition, triggering an abrupt starburst. We implement a `jeans_bypass_holdback` option that allows the Jeans criterion to operate independently of the holdback, with a configurable level limit (`n_jeans_bypass_levels`) to prevent uncontrolled refinement cascades. We discuss the interplay between the Jeans criterion, the polytropic equation of state, and the density threshold `d_jeans_thre`, showing that the polytropic temperature floor provides a natural self-regulation mechanism for the bypass.

1 AMR Refinement in ramses

In RAMSES, each cell at level $\ell < N_{\mathrm{levelmax}}$ is tested for refinement at every coarse time-step. The decision to subdivide a cell into $2^{N_{\mathrm{dim}}}$ children is governed by a combination of physical, geometric, and cosmological criteria.

1.1 Refinement logic

The overall refinement flag for a cell is determined by the following Boolean expression:

$$\mathcal{R} = \neg\,\mathcal{H} \;\wedge\; \mathcal{G} \;\wedge\; \big(\mathcal{M} \vee \mathcal{S} \vee \mathcal{D} \vee \mathcal{P} \vee \mathcal{U} \vee \mathcal{J}\big), \tag{1}$$

where each symbol is defined below.

1.1.1 Physical criteria (OR)

1. **Quasi-Lagrangian mass criterion ($\mathcal{M}$).** The cell is flagged if its gas density exceeds a level-dependent threshold:

$$\rho_{\rm cell} \;\geq\; m_{\rm refine}(\ell)\,\frac{m_{\rm sph}}{V_{\rm cell}}, \tag{2}$$

   where $m_{\rm sph}$ is the effective SPH kernel mass and $V_{\rm cell} = (\Delta x_\ell)^3$. This criterion ensures that regions containing roughly $m_{\rm refine}$ or more particles per cell are refined, providing a quasi-Lagrangian resolution that tracks gravitational collapse. Controlled by the namelist array `m_refine(1:nlevelmax)`.

2. **Sink particle forcing ($\mathcal{S}$).** If `sink_refine=.true.`, any cell containing a sink cloud particle (identified by `idp` < 0 and `tp` $= 0$) is unconditionally flagged up to $N_{\rm levelmax}$, guaranteeing maximum resolution around each accretion region.

3. **Density gradient ($\mathcal{D}$).** The normalised density gradient between a cell and each of its $2N_{\rm dim}$ face-neighbours is computed as

$$\epsilon_\rho = 2\,\max\!\left(\frac{|\rho_+ - \rho_0|}{|\rho_+ + \rho_0| + \rho_{\rm floor}},\; \frac{|\rho_0 - \rho_-|}{|\rho_0 + \rho_-| + \rho_{\rm floor}}\right). \tag{3}$$

   If $\epsilon_\rho >$ `err_grad_d` along any axis, the cell is flagged. This captures density discontinuities such as shocks and contact waves.

4. **Pressure gradient ($\mathcal{P}$).** Analogous to the density gradient but applied to the thermal pressure:

$$\epsilon_P = 2\,\max\!\left(\frac{|P_+ - P_0|}{|P_+ + P_0| + P_{\rm floor}},\; \frac{|P_0 - P_-|}{|P_0 + P_-| + P_{\rm floor}}\right), \tag{4}$$

   flagging when $\epsilon_P >$ `err_grad_p`.

5. **Velocity gradient ($\mathcal{U}$).** A normalised velocity jump relative to the local sound speed:

$$\epsilon_v = 2\,\max\!\left(\frac{|v_+ - v_0|}{c_{s,+} + c_{s,0} + |v_+| + |v_0| + v_{\rm floor}},\; \frac{|v_0 - v_-|}{c_{s,0} + c_{s,-} + |v_0| + |v_-| + v_{\rm floor}}\right), \tag{5}$$

   flagging when $\epsilon_v >$ `err_grad_u` in any component. This criterion targets strong velocity shears and accretion shocks.

6. **Jeans length ($\mathcal{J}$).** When `jeans_refine(`ℓ`)` $= N_J > 0$, the cell is flagged if the local Jeans length is shorter than N_J cell widths:

$$\lambda_{\rm J} = \sqrt{\frac{\pi\, c_s^2}{G\,\rho}} < N_J\,\Delta x_\ell, \tag{6}$$

implementing the **?** condition to prevent artificial fragmentation in self-gravitating gas.

An additional density threshold `d_jeans_thre` restricts this criterion to cells with $n_{\mathrm{H}} \geq$ `d_jeans_thre`.

1.1.2 Geometry mask (AND)

Ellipsoidal region ($\mathcal{G}$). When `r_refine(`ℓ`)` > -1, only cells inside a generalised ellipsoidal region are permitted to refine. The test is

$$\left(\frac{2|\Delta x|}{r}\right)^{e_r} + \left(\frac{2|\Delta y|}{a\,r}\right)^{e_r} + \left(\frac{2|\Delta z|}{b\,r}\right)^{e_r} < 1, \tag{7}$$

where $r =$ `r_refine` is the region diameter, $a =$ `a_refine` and $b =$ `b_refine` are the axis ratios, and $e_r =$ `exp_refine` is the norm exponent ($e_r = 2$ gives an ellipsoid; $e_r \to \infty$ a rectangular box). Cells outside this region are *un-flagged* regardless of physical criteria, restricting high resolution to a zoom-in target.

1.1.3 Cosmological holdback (NOT)

Resolution holdback ($\mathcal{H}$). In cosmological runs with cooling (`cosmo=.true.`, `cooling=.true.`), an overly eager refinement at high redshift can exhaust memory before the epoch of interest. When `q_refine_holdback=.true.` and the physical cell size $\Delta x_\ell < 4^{1/N_{\mathrm{dim}}}\,\Delta x_{N_{\mathrm{levelmax}}}/a_{\mathrm{exp}}$ for levels $\ell >$ `nlevelmax_part` $+ 3$, the *entire* level is blocked from further refinement:

$$\mathcal{H}(\ell) = \big[\ell > \ell_{\mathrm{part}} + \ell_{\mathrm{collapse}}\big] \;\wedge\; \big[\Delta x_\ell < 4^{1/N_{\mathrm{dim}}}\,\Delta x_{\mathrm{min}}/a\big], \tag{8}$$

where ℓ_{part} is the particle initial-condition level (typically ℓ_{min}) and $\ell_{\mathrm{collapse}} = 3$. This enforces a constant physical resolution at low redshift and a constant comoving resolution at high redshift, gradually releasing deeper levels as the universe expands.

The first holdback level ℓ_{hb} is the smallest ℓ satisfying Eq. (**??**):

$$\ell_{\mathrm{hb}} = \min\big\{\ell > \ell_{\mathrm{part}} + \ell_{\mathrm{collapse}} \;:\; \Delta x_\ell < 4^{1/N_{\mathrm{dim}}}\,\Delta x_{\mathrm{min}}/a\big\}. \tag{9}$$

1.2 Buffer zones

After applying equation (**??**), flagged cells are expanded by `nexpand(`ℓ`)` cell widths in each direction via a dilation operator (`smooth_fine`). This cubic buffer ensures smooth refinement boundaries, preventing the Godunov solver from encountering abrupt level transitions that would degrade numerical accuracy.

2 The Starburst Problem

The holdback mechanism blocks *all* refinement criteria at levels $\ell \geq \ell_{\mathrm{hb}}$, including the Jeans criterion. This creates two related problems:

Artificial fragmentation. The Jeans length criterion $\mathcal{J}$ differs fundamentally from the other refinement triggers: it is a *numerical stability* requirement (**?**), not merely a resolution preference. Violating the Jeans condition leads to artificial fragmentation — a qualitatively wrong result — regardless of epoch. Yet in the standard RAMSES logic (equation **??**), the cosmological holdback $\mathcal{H}$ blocks $\mathcal{J}$ uniformly with all other criteria. At high redshift, dense self-gravitating regions may therefore be held at insufficient resolution, producing spurious fragmentation that propagates into lower-redshift structure.

Abrupt SFR transition. When the holdback releases level ℓ (i.e. $\mathcal{H}(\ell)$ transitions from true to false), all accumulated criteria fire simultaneously. Many cells that have been gravitationally collapsing during the holdback period now suddenly satisfy the refinement condition, leading to a burst of new cells and an abrupt spike in the star formation rate.

3 Jeans Bypass: Design and Implementation

3.1 Modified refinement logic

The key observation is that the Jeans length criterion is *cell-local*: it depends only on the density and temperature of the cell itself (`uold`), with no neighbor information required. This allows a lightweight evaluation even when the full refinement machinery (which requires neighbor gathering for $\mathcal{D}$, $\mathcal{P}$, $\mathcal{U}$) is skipped by the holdback.

With `jeans_bypass_holdback=.true.`, the refinement logic becomes

$$\mathcal{R}' = \Big[\neg\,\mathcal{H} \wedge \mathcal{G} \wedge \big(\mathcal{M}\vee\mathcal{S}\vee\mathcal{D}\vee\mathcal{P}\vee\mathcal{U}\vee\mathcal{J}\big)\Big] \vee \Big[\mathcal{H} \wedge \mathcal{J} \wedge (\ell < \ell_{\mathrm{hb}}+n_{\mathrm{bypass}})\Big], \tag{10}$$

where $n_{\mathrm{bypass}} \equiv$ `n_jeans_bypass_levels` (default 1). The geometry mask $\mathcal{G}$ still applies through the non-holdback branch, preserving zoom-in spatial restrictions. Only the temporal holdback $\mathcal{H}$ is bypassed for $\mathcal{J}$, and only within a limited number of levels.

This modification is analogous to the existing treatment of sink particles: $\mathcal{S}$ is already evaluated inside the holdback-protected block in the current code, but unlike $\mathcal{J}$ it only triggers in cells that already host a sink. Extending the same exemption to $\mathcal{J}$ ensures numerical integrity for gas that is *about to* form a sink.

3.2 Level limitation: preventing cascades

Without a level limit ($n_{\rm bypass} \to \infty$), the Jeans bypass creates a catastrophic cascade. When a cell at the holdback level $\ell_{\rm hb}$ is flagged by the Jeans criterion, child cells are created at level $\ell_{\rm hb} + 1$. Since level $\ell_{\rm hb} + 1$ is also in the holdback zone, the bypass applies again, flagging cells at $\ell_{\rm hb} + 2$, and so on until $\ell_{\rm max}$. This rapidly exhausts available memory.

The cascade occurs because the Jeans length in radiatively cooling gas (below the star-formation density threshold) shrinks without bound as the temperature drops. There is no physical mechanism to halt the Jeans criterion from triggering at every successive level — only the polytropic temperature floor (Section **??**) or a hard level limit can stop it.

Table **??** shows the progression observed in a test run without level limit:

Table 1: Cascade refinement without level limit ($a = 0.164$, $\ell_{\rm hb} = 14$, $\ell_{\rm max} = 17$). All levels ≥ 14 are in the holdback zone.

Level	$\Delta x/\Delta x_{\rm min}$	Holdback	Bypass	$\mathcal{J}$ fires	Result
≤ 12	≥ 32	No	—	Yes (normal)	Normal
13	16	No	—	Yes (normal)	Normal
14	8	Yes	Active	Yes	$\to$ lv. 15 created
15	4	Yes	Active	Yes	$\to$ lv. 16 created
16	2	Yes	Active	Yes	$\to$ lv. 17 created
17	1	$\ell_{\rm max}$	—	—	OOM crash

The memory footprint grew from 81% to 88.5% in a single coarse step before crashing (Job 122644). The fine-step timestep also decreased by 8× due to subcycling at levels 15–17.

With $n_{\rm bypass} = 1$, only level $\ell_{\rm hb} = 14$ is bypassed, creating cells at level 15 but no further. This provides a controlled pre-refinement before the holdback releases.

4 Role of the Polytropic Temperature Floor

4.1 Jeans length in the polytropic regime

In RAMSES, the star-formation module enforces a polytropic equation of state above the density threshold $n_\star$:

$$T(n) = T_{2,\star} \left(\frac{n}{n_\star}\right)^{g_\star - 1}, \qquad n \geq n_\star, \tag{11}$$

with default parameters $T_{2,\star} = 750\,{\rm K}$, $n_\star = 0.1\,{\rm cm}^{-3}$, $g_\star = 5/3$. This floor is applied in the cooling routine and is already reflected in `uold`, so the Jeans

length computation in `jeans_length_refine()` automatically includes the polytropic stabilisation.

Substituting Eq. (**??**) into Eq. (**??**):

$$\lambda_{\rm J} \propto \sqrt{\frac{T}{\rho}} \propto \sqrt{\rho^{g_\star - 1}/\rho} = \rho^{(g_\star - 2)/2}. \tag{12}$$

For $g_\star = 5/3$: $\lambda_{\rm J} \propto \rho^{-1/6}$. The Jeans length *decreases very slowly* with density — one or two additional levels of refinement are typically sufficient to resolve it.

In contrast, for gas below $n_\star$ (no polytropic floor), radiative cooling can drive T to very low values while ρ remains moderate. This yields $\lambda_{\rm J} \ll \Delta x_\ell$ at many levels, causing the cascade.

4.2 Self-regulation via `d_jeans_thre`

The density threshold parameter `d_jeans_thre` (in units of $n_{\rm H}\,[{\rm cm}^{-3}]$) restricts the Jeans criterion to cells above a given density:

$$\mathcal{J}(\text{cell}) = \begin{cases} N_J\,\Delta x_\ell \geq \lambda_{\rm J} & \text{if } n \geq \texttt{d_jeans_thre}, \\ \text{false} & \text{otherwise.} \end{cases} \tag{13}$$

Setting `d_jeans_thre` $\approx n_\star$ restricts the Jeans criterion to the polytropic regime. The cascade then self-limits through the following mechanism:

1. At level $\ell_{\rm hb}$: only cells with $n \geq n_\star$ are checked. The polytropic floor guarantees $T \geq T_{2,\star}$, so $\lambda_{\rm J}$ is relatively large.
2. If $N_J\,\Delta x_{\ell_{\rm hb}} > \lambda_{\rm J}$, the cell is flagged and children at $\ell_{\rm hb}+1$ are created.
3. At level $\ell_{\rm hb}+1$: Δx is halved, density is approximately unchanged (refinement preserves cell-averaged quantities), $\lambda_{\rm J}$ is approximately unchanged.
4. $N_J\,\Delta x_{\ell_{\rm hb}+1} = N_J\,\Delta x_{\ell_{\rm hb}}/2 < \lambda_{\rm J} \Rightarrow$ Jeans criterion is *satisfied* (the Jeans length is now resolved) $\Rightarrow$ no further flagging $\Rightarrow$ cascade stops naturally.

This provides a *physics-based* self-regulation, complementary to the *hard limit* imposed by `n_jeans_bypass_levels`.

The two regulators are compared in Table **??**.

When `d_jeans_thre` $= 0$ (default), *all* cells are checked, including low-density gas where cooling produces arbitrarily small $\lambda_{\rm J}$. This is the root cause of the unlimited cascade observed in Section **??**.

Table 2: Two complementary mechanisms for preventing the bypass cascade.

Mechanism	Type	Behaviour
`n_jeans_bypass_levels`	Hard limit	Caps extra levels regardless of physics
`d_jeans_thre` $\approx n_\star$	Physics-based	Polytropic floor limits $\lambda_{\rm J}$ decrease

5 Implementation Details

5.1 Code structure

The implementation modifies two source files (plus parameter declaration and namelist reading).

`flag_utils.kjhan.f90` — `userflag_fine()` The holdback early-return is modified from a simple `return` to a conditional block:

```
if(prevent_refine .and. q_refine_holdback) then
   if(jeans_bypass_holdback .and. hydro &
      .and. n_jeans_bypass_levels > 0) then
      ! Find the first holdback level
      dx_min = (0.5D0**nlevelmax)*scale
      holdback_threshold = (4d0**(1d0/ndim))*(dx_min/aexp
         )
      holdback_first = nlevelmax + 1
      do jlevel = nlevelmax_part+nlevel_collapse+1,
         nlevelmax
         dx_test = (0.5d0**jlevel)*scale
         if(dx_test < holdback_threshold) then
            holdback_first = jlevel
            exit
         endif
      end do
      ! Only bypass within allowed range
      if(ilevel < holdback_first &
         + n_jeans_bypass_levels) then
         call jeans_only_flag(ilevel)
      endif
   endif
   return
endif
```

`hydro_flag.kjhan.f90` — `jeans_only_flag()` A new lightweight subroutine that:

1. Loops over active grids with OpenMP parallelism.
2. Computes cell indices without neighbor gathering.

3. Calls the existing `jeans_length_refine()` subroutine (no code duplication).

4. Sets `flag1` for flagged cells and updates `nflag`.

This is possible because `jeans_length_refine()` only accesses cell-local data (`uold(ind_cell,...)`) and requires no neighbor information, unlike the gradient-based criteria $\mathcal{D}$, $\mathcal{P}$, $\mathcal{U}$.

5.2 Namelist parameters

In `amr_parameters.jaehyun.f90`:

```
logical :: jeans_bypass_holdback = .false.
integer :: n_jeans_bypass_levels = 1
```

Both are read via `&REFINE_PARAMS` in `read_hydro_params.f90`.

5.3 Safety properties

- **Default off**: With `jeans_bypass_holdback=.false.` (default), the new code path is never entered. Results are bit-identical to the unmodified code.
- **No neighbor access**: `jeans_only_flag()` is safe to call at any point in the refinement cycle.
- **OMP safety**: `flag1` writes are grid-partitioned (no race conditions); `nflag` uses reduction.
- **Hard level limit**: `n_jeans_bypass_levels` provides a fail-safe even if the physics-based regulation (`d_jeans_thre`) is misconfigured.

6 Summary of Parameters

Table **??** lists all refinement-related namelist parameters, including the new bypass parameters.

The recommended configuration for cosmological runs with star formation is:

```
&REFINE_PARAMS
m_refine           = 9*8.,
jeans_refine       = 9*4.,
jeans_bypass_holdback  = .true.
n_jeans_bypass_levels  = 1
d_jeans_thre       = 0.1          ! = n_star
q_refine_holdback = .true.
/
```

Table 3: AMR refinement parameters in the `&AMR_PARAMS` and `&REFINE_PARAMS` namelists. New parameters are marked with $\star$.

Parameter	Default	Role
`m_refine`	8.0	Lagrangian mass threshold (per level)
`sink_refine`	F	Force max level around sinks
`err_grad_d`	-1	Density gradient threshold
`err_grad_p`	-1	Pressure gradient threshold
`err_grad_u`	-1	Velocity gradient threshold
`jeans_refine`	-1	Jeans length in cells (per level)
`d_jeans_thre`	0	Density threshold for Jeans ($n_{\rm H}\,{\rm cm}^{-3}$)
`r_refine`	-1	Geometry region diameter (per level)
`a_refine`	1.0	Ellipticity Y/X
`b_refine`	1.0	Ellipticity Z/X
`exp_refine`	2.0	Norm exponent
`x/y/z_refine`	0.5	Region centre
`nexpand`	1	Buffer zone width (per level)
`q_refine_holdback`	F	Cosmological holdback
$\star$ `jeans_bypass_holdback`	F	Enable Jeans bypass of holdback
$\star$ `n_jeans_bypass_levels`	1	Max extra levels beyond $\ell_{\rm hb}$

7 Test Results

All tests use the HR5 medium-box setup: $\ell_{\rm min} = 9$, $\ell_{\rm max} = 17$, 12 MPI ranks, OMP=5, restarting from output 5 ($a = 0.164$, $z \approx 5.1$, nstep_coarse=241).

7.1 Bit-identical verification

With `jeans_bypass_holdback=.false.` (default) and no `jeans_refine`, the new binary produces results identical to the reference run: econs=6.13×10^{-4} at step 242, econs=6.71×10^{-4} at step 243 (Job 122643).

7.2 Unlimited bypass ($n_{\rm bypass} = \infty$, `d_jeans_thre` $= 0$)

With `jeans_refine=9*4.`, `jeans_bypass_holdback=.true.`, and no level limit: cascade refinement from level 14 through 17 within a single coarse step. Memory grew from 81% to 88.5% before OOM crash at fine step 1580. Fine-step timestep decreased by 8× due to subcycling (Job 122644).

7.3 Limited bypass ($n_{\rm bypass} = 1$, `d_jeans_thre` $= 0$)

With `n_jeans_bypass_levels=1`: only level 14 is bypassed, cells at level 15 may be created but no further cascade. Results: Job 122663 (pending).

7.4 Future: `d_jeans_thre` $= n_\star$

A test with `d_jeans_thre=0.1` will verify the physics-based self-regulation, potentially allowing $n_{\text{bypass}} > 1$ without cascade, since only gas in the polytropic regime would trigger the Jeans criterion.

8 Discussion

8.1 Physical motivation

The holdback mechanism was designed to prevent over-refinement from the mass criterion at early times. However, it inadvertently blocks the Jeans criterion, which has a different physical motivation: resolving the Jeans length prevents artificial fragmentation and ensures numerically converged star formation.

By allowing the Jeans criterion to operate ahead of the holdback release, we expect:

- Dense gas that will form stars is pre-refined before the holdback releases.
- When the holdback releases, the transition is smoother because the densest regions are already at the appropriate resolution.
- The SFR curve shows a gradual onset rather than an abrupt step.

8.2 Interplay of the three parameters

The bypass is controlled by three parameters that form a layered defence:

Layer 1: `jeans_bypass_holdback` (on/off) Master switch. When off (default), the system is identical to standard RAMSES.

Layer 2: `d_jeans_thre` (physics-based regulation) Restricts the Jeans check to dense gas where the polytropic floor operates. In this regime, $\lambda_{\rm J} \propto \rho^{-1/6}$ varies slowly, and 1–2 extra refinement levels suffice to resolve it. Provides self-consistent, physics-driven cascade termination.

Layer 3: `n_jeans_bypass_levels` (hard safety limit) Absolute cap on the number of extra levels. Prevents OOM regardless of physics configuration. Essential when `d_jeans_thre=0` (all cells checked).

References

Truelove, J. K., et al., 1997, ApJ, 489, L179

Dubois, Y., et al., 2021, A&A, 651, A109